普物公式整理

前面没空做了

定理

Chapter28 电势能、电势

定义

1. $\Delta U_{a\to b}=-\int_a^b \vec{F}\cdot d\vec{l}$，若定义 $U_\infty=0$ 则 $U_p=\int_p^\infty \vec{F}\cdot d\vec{l}$
    - 单位 : J
2. 力矩 $\vec{\tau}=\vec{p}\times\vec{E}$
    - 由此定义的 $U=-\vec{p}\cdot\vec{E}$
3. $V=\frac{U}{q_0}$ 若定义 $V_\infty=0$ 则 $V_p=\int_p^\infty \vec{E}\cdot d\vec{l}$
    - 单位 : V = J/C

从 E 到 V : $V_p=\int_p^\infty \vec{E}\cdot d\vec{l}$

从 V 到 E : $E=-\nabla V$

定理

1. **静电场环路定理**

$$\oint \vec{E}\cdot\vec{dl}=0$$

$$\nabla\times\vec{E}=0$$

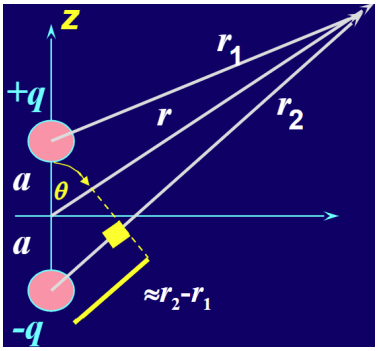
注：对于变化的电场，有（参见Chapter34)

$$\oint \vec{E}\cdot\vec{dl}=-\frac{d\Phi_B}{dt}$$

$$\nabla\times\vec{E}=-\frac{\partial\vec{B}}{\partial t}$$

模型

1. 点电荷
    - $U(r)=\frac{q_1q_2}{4\pi\epsilon_0 r}$
    - $V(r)=\frac{q}{4\pi\epsilon_0 r}$
2. 电偶极矩

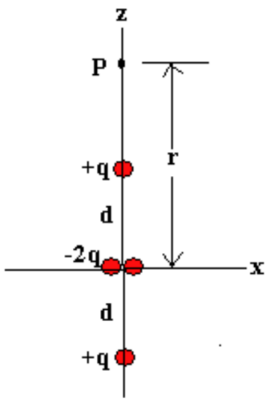


$$V(r)=\frac{q}{4\pi\epsilon_0}(\frac{1}{r_1}-\frac{1}{r_2})\approx\frac{p\cdot\hat{r}}{4\pi\epsilon_0 r^2}$$

如果使用 $r-\theta$ 的坐标系

$$V(r,\theta)=\frac{2aq\cos\theta}{4\pi\epsilon_0 r^2}\Rightarrow\vec{E}=\frac{p}{4\pi\epsilon_0 r^3}((2\cos\theta)\hat{r}+(\sin\theta)\hat{\theta})$$

3. 电四偶极矩



$$V(r)=\frac{1}{4\pi\epsilon_0}\frac{Q}{r^3(1-d^2/r^2)}$$
$$Q=2qd^2(Electric\ quadrupole\ moment)$$

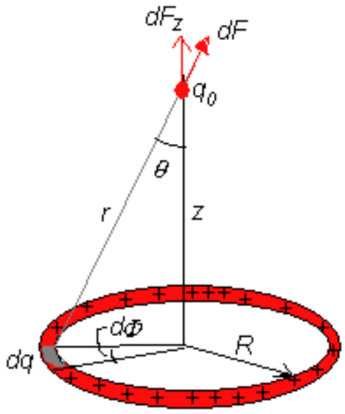
4. 带电球壳

$$(Gauss'\ Law\Rightarrow)E=\begin{cases}\frac{q}{4\pi\epsilon_0 r^2},r>R\\0,r\le R\end{cases}\Rightarrow V(r)=\begin{cases}\frac{q}{4\pi\epsilon_0 r},r>R\\\frac{q}{4\pi\epsilon_0 R},r\le R\end{cases}$$

电势能：微元

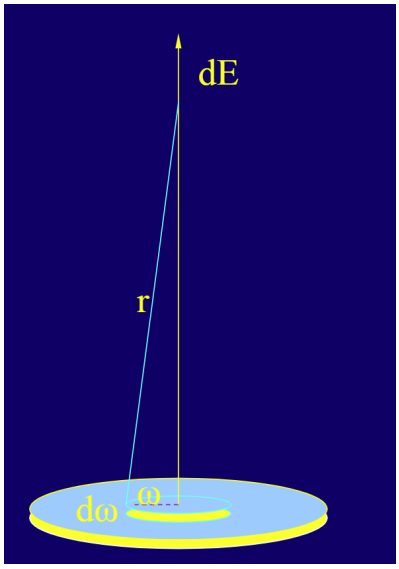
$$\begin{aligned}U&=\sum_{1\le i<j\le n}\frac{q_iq_j}{4\pi\epsilon_0 r_{ij}}=\frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\frac{q_iq_j}{4\pi\epsilon_0 r_{ij}}\\&=\frac{1}{2}\sum_{i=1}^{n}q_iV_i=\frac{1}{2}\int Vdq=\frac{1}{2}\frac{q}{4\pi\epsilon_0 R}q\\&=\frac{q^2}{8\pi\epsilon_0 R}\end{aligned}$$

5. 带电圆环



$$V=\int dV=\frac{1}{4\pi\epsilon_0}\oint\frac{\lambda ds}{r}=\frac{q}{4\pi\epsilon_0}\frac{1}{\sqrt{z^2+R^2}}$$

6. 带电圆盘



$$\begin{aligned}V&=\int_0^R\frac{dq}{4\pi\epsilon_0}\frac{1}{\sqrt{z^2+w^2}}\\&=\int_0^R\frac{2\pi wdw\sigma}{4\pi\epsilon_0}\frac{1}{\sqrt{z^2+w^2}}\\&=\frac{\sigma}{2\epsilon_0}(\sqrt{z^2+R^2}-z)\end{aligned}$$

$$if(z>>R)\ V=\frac{q}{4\pi\epsilon_0 z}$$

$$E=-\nabla V=-\frac{\partial V}{\partial z}=-\frac{\sigma}{2\epsilon_0}(1-\frac{1}{\sqrt{1+R^2/z^2}})$$

其他

1. 尖端放电

   在电势相同的情况下，尖端附近电场更强，容易放电

2. 镜像法处理感应电荷分布

   感应电荷可以等效成一个导体内的镜像电荷，从而达到在导体表面产生电场与源电荷电场相抵消的效果

Chapter29(30) 电容(capacitance)和电介质(dielectrics)

定义

1. 电容 $C=\frac{q}{\Delta V}$
2. 电容的串并联
   - 串联：$C=C_1+C_2$
   - 并联：$\frac{1}{C}=\frac{1}{C_1}+\frac{1}{C_2}$
3. 电场能量密度（单位体积电场中的能量）$u_E=\frac{1}{2}\epsilon_0E^2$

   推导（以平行板为例）：
   - 给电容器充电 $U=\frac{1}{C}\int_0^Q qdq=\frac{1}{2}\frac{Q^2}{C}$ 同时也 $U=\frac{1}{2}CV^2$
   - $U=\frac{1}{2}\frac{Q^2}{A\epsilon_0/d}$ 且 $E=\frac{\sigma}{\epsilon_0}=\frac{Q}{\epsilon_0A}\Rightarrow U=\frac{1}{2}E^2\epsilon_0Ad$
   - $u_E=\frac{W}{volume}=\frac{U}{Ad}=\frac{1}{2}\epsilon_0E^2$
   - ☐ 例题：验证该 u_E 适用于圆筒形电容器（tip: 即将用 $U=\int u_Edv$ 得到的结果与预期进行比较）
4. 介电常数 κ_e： $C=\kappa_eC_0$ 介电常数表现了电容器中使用该种介质与真空相比的电容差距（总是大于1）

   微观理解
   - 对于无极分子电介质，初始 $p=qd=0$，在 $\vec{E}_0$ 作用下 $p\neq0$ 从而在两端产生正负电荷。产生束缚电荷，反向的电场。“电子位移极化”
   - 对于有极分子电介质，初始 p 方向混乱，求和为 0，在 $\vec{E}_0$ 作用下方向统一从而在两端产生正负电荷。“取向极化”（当然也存在电子位移极化，然而相比之下效应太小，仅在电场高频变化的时候比较明显）
5. 极化强度矢量 $\vec{P}=\frac{\sum\vec{p}_i}{\Delta V}$
   - **介质表面的**σ : $\sigma=\vec{P}\cdot\vec{n}$ （$\vec{n}$ 是法向量）
   - 计算思路 $P\Rightarrow\sigma_e'\Rightarrow E'\Rightarrow E$ （E'是在理解过程中认定介质产生的束缚电荷的电场，名为退极化场depolarization field，E 是最后表现出来的电场）
   - 可用的一个式子 $\oiint\vec{P}\cdot d\vec{A}=-\sum q_{in}$
6. 极化率 χ_e : 对于一般的具有各向同性的材料，有 $P=\chi_e\epsilon_0E$
   - $\kappa_e=1+\chi_e$
7. 电位移矢量（电感应强度）$\vec{D}=\epsilon_0E+P$

定理

1. **介质中的高斯定律**

$$\oiint D\cdot dA=\sum q_{in}$$

note:此处的 q_{in} 指的是内部的自由电荷（也即束缚电荷不算在内）

2. 环路定律不成立

$$\oint\vec{D}\cdot\vec{dl}\neq0$$

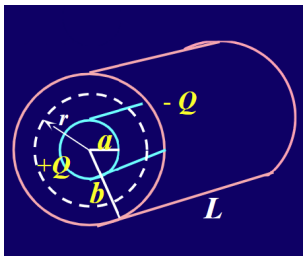
因为 κ_e 在不同路径不一样

模型

1. 平行板电容器

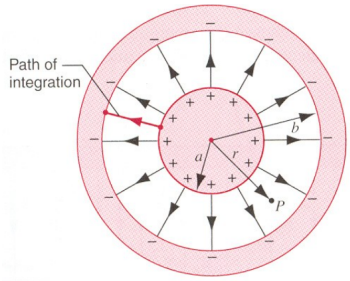
$$C=\frac{q}{\Delta V}=\frac{\sigma A}{\frac{\sigma d}{\kappa_e\epsilon_0}}=\kappa_e\frac{\epsilon_0 A}{d}$$

2. 圆筒形电容器



$$C=\frac{q}{\Delta V}=\frac{\kappa_e Q}{\int_a^b\frac{Q}{2\pi\epsilon_0 rL}dr}=\kappa_e\frac{2\pi\epsilon_0 L}{\ln\frac{b}{a}}$$

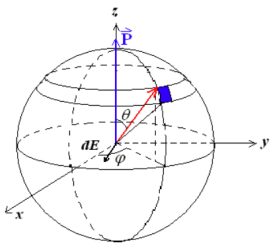
3. 球形电容器



$$C=\frac{q}{\Delta V}=\kappa_e\frac{q}{\int_a^b\frac{q}{4\pi\epsilon_0 r^2}dr}=\kappa_e\frac{4\pi\epsilon_0 ab}{b-a}$$

令 $b\to\infty$，单独一个球也是电容器，$C=4\kappa_e\pi\epsilon_0 a$

4. 球体介质在电场中的退极化场



取表面的微元（蓝色部分）

$$\begin{aligned}
\sigma_e'&=P\cos\theta\\
dE'&=\frac{dq'}{4\pi\epsilon_0 R^2}=\frac{\sigma_e' dA}{4\pi\epsilon_0 R^2}=\frac{P\cos\theta dA}{4\pi\epsilon_0 R^2}\\
dA&=Rd\theta R\sin\theta d\varphi\\
dE_z'&=-dE'\cos\theta=\frac{P\cos\theta\sin\theta d\theta d\varphi}{4\pi\epsilon_0}\\
E_z'&=\oiint dE_z'=-\frac{P}{3\epsilon_0}
\end{aligned}$$

Chapter31 恒定电流

定义

1. 电流强度 $i = \frac{dq}{dt}$
2. 电流密度矢量 j $di = \vec{j} \cdot \vec{dA}$
   - $\oiint_A \vec{j} \cdot \vec{dA} = -\frac{dq}{dt}$
3. 恒定电流的电流场有 $\oiint_A \vec{j} \cdot \vec{dA} = 0$
4. 电阻率和电导率
   - $R = \rho \frac{L}{A}$
   - ρ 电阻率， $\sigma = \frac{1}{\rho}$ 电导率
   - 对于金属在一定范围内有 $\rho(T) = \rho_0 + \alpha T$
5. 电功率 $P = \frac{W}{\Delta t}$
6. 电动势 $\epsilon = \int_-^+ \vec{K} \cdot \vec{dl}$
   - 把单位正电荷从负极由电源内部移到正极的非静电力做功

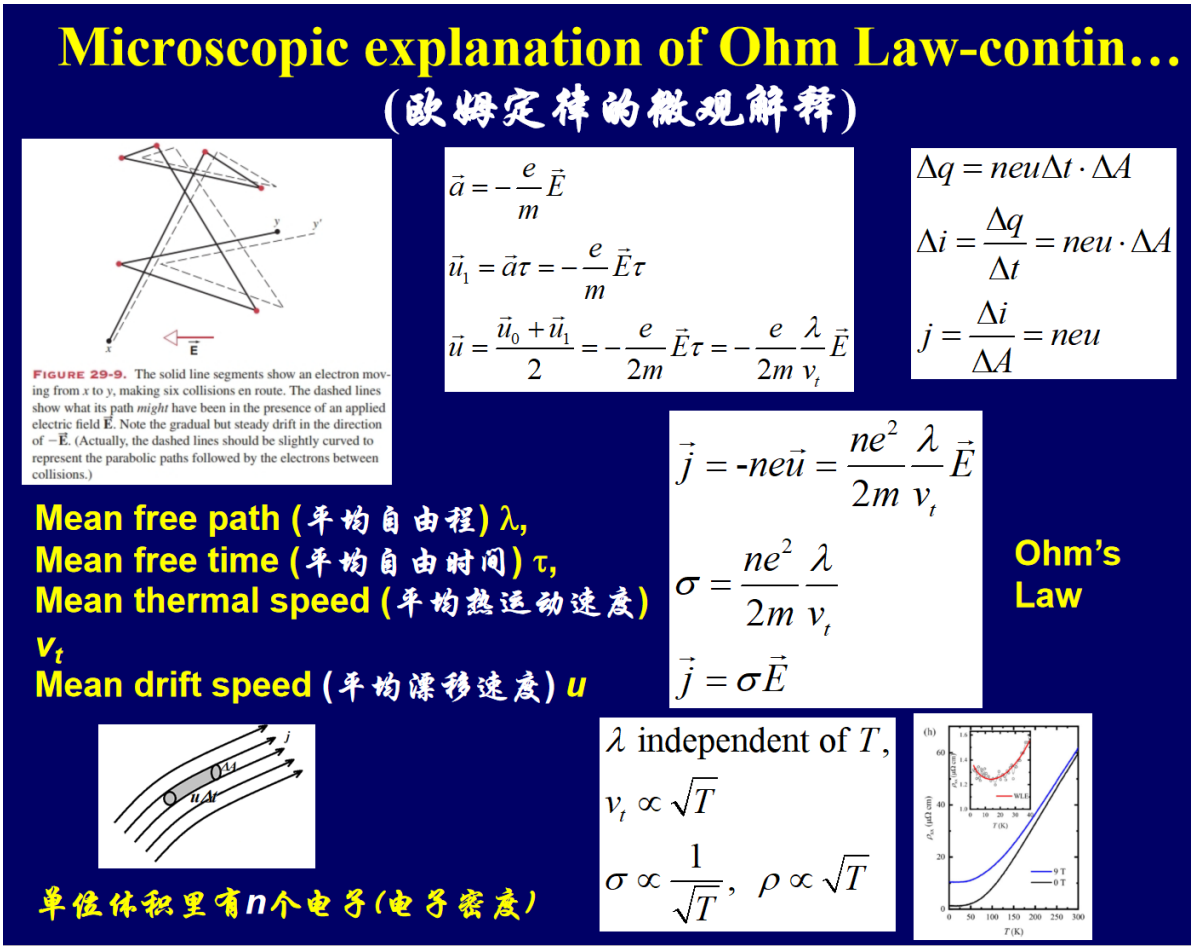
定理

1. 欧姆定律 $R = \frac{dV}{dI}$
   - 微分形式： $\vec{j} = \sigma \vec{E}$

$$\Delta i = \frac{\Delta V}{R} \Rightarrow j\Delta A = \frac{E\Delta l}{\rho \frac{\Delta l}{\Delta A}}$$

$$j = \frac{E}{\rho} = \sigma E$$

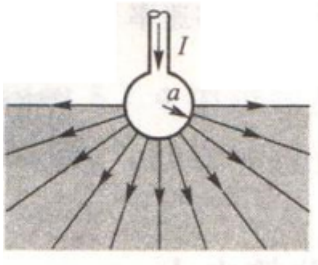
   - 微观解释



2. 焦耳定律 $P = i^2R = \frac{V^2}{R}$

模型

1. 电极的接地电阻



视为多层球壳作电阻

$$R = \int \rho \frac{dl}{A} = \int_a^\infty \rho \frac{dr}{2\pi r^2} = \frac{\rho}{2\pi a}$$

Chapter 32(33) 稳恒磁场

定义

1. 磁感应强度 $\vec{B} = \frac{\mu_0}{4\pi} \oint \frac{i_1 \vec{ds_1} \times \hat{r_{12}}}{r_{12}^2}$
   - 单位 $1T = 1N/(m \cdot A)$
2. 磁力 $\vec{dF} = i_2 \vec{ds_2} \times \vec{B}$
3. 磁偶极矩 $\vec{\mu_m} = iA\vec{n}$
   - 如果多匝线圈要乘以相应的线圈匝数
   - 方向为 A 的法向方向（$\vec{n}$）
4. 力矩 $\vec{\tau} = iA(\vec{n} \times \vec{B}) = \vec{\mu_m} \times \vec{B}$
   - 由此定义的 $U = -\vec{\mu_m} \cdot \vec{B}$ （貌似都是 - X偶极矩 点乘 X场）
5. 洛伦兹力 $\vec{f} = q\vec{v} \times \vec{B}$
   - $\vec{F} = i\vec{ds} \times \vec{B}$ 对应的微观描述

定理

1. Biot-Savart Law

   其实就是 $\vec{B} = \frac{\mu_0}{4\pi} \oint \frac{i_1 \vec{ds_1} \times \hat{r_{12}}}{r_{12}^2}$
2. 高斯定律

$$\oiint \vec{B} \cdot \vec{dA} = 0$$

$$\nabla \cdot \vec{B} = 0$$

3. 磁场安培环路定律

$$\oint \vec{B} \cdot \vec{dl} = \mu_0 \sum_{in\ loop} i$$

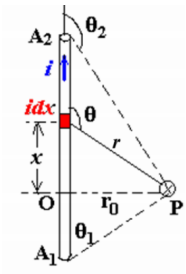
notes

- i 的符号： 符合右手定则为正，反之为负

- 如果多次穿入穿出则要多次计算

模型

1. 长直导线



$$B=\frac{\mu_0 i}{4\pi r_0}(\cos\theta_1-\cos\theta_2)$$

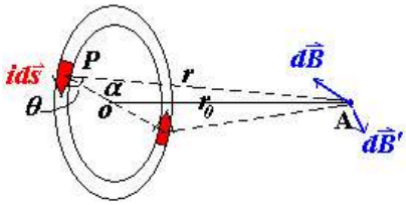
无穷长导线则有（可用安培环路）

$$B=\frac{\mu_0 i}{2\pi r_0}$$

补充：导线内部：

$$B=\frac{\mu_0 i r}{2\pi R^2}$$

1. 环形导线



$$B=\frac{\mu_0}{2}\frac{iR^2}{(R^2+r_0^2)^{3/2}}=\frac{\mu_0}{2\pi}\frac{\mu}{(R^2+r_0^2)^{3/2}}$$

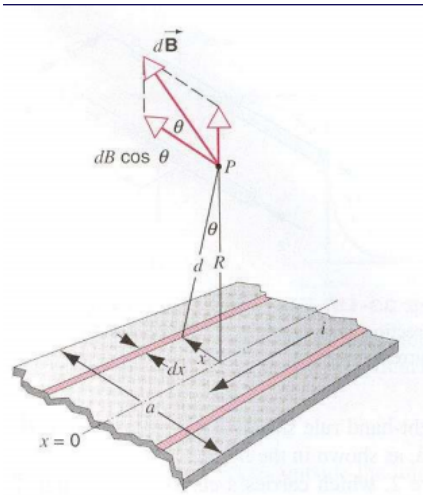
若在导线正中则有

$$B=\frac{\mu_0 i}{2R}$$

若距离很远则有

$$B=\frac{\mu_0 i R^2}{2r_0^3}$$

3. 大平板



$$B_x = \frac{\mu_0 i}{\pi a}\arctan\frac{a}{2R}$$

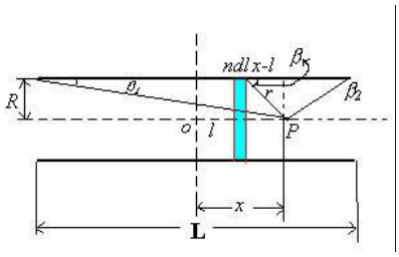
若离平板很近则有

$$B = \frac{\mu_0 i}{2\pi R}$$

若离平板很远则有

$$B = \frac{\mu_0 i}{2a}$$

4. 螺旋管

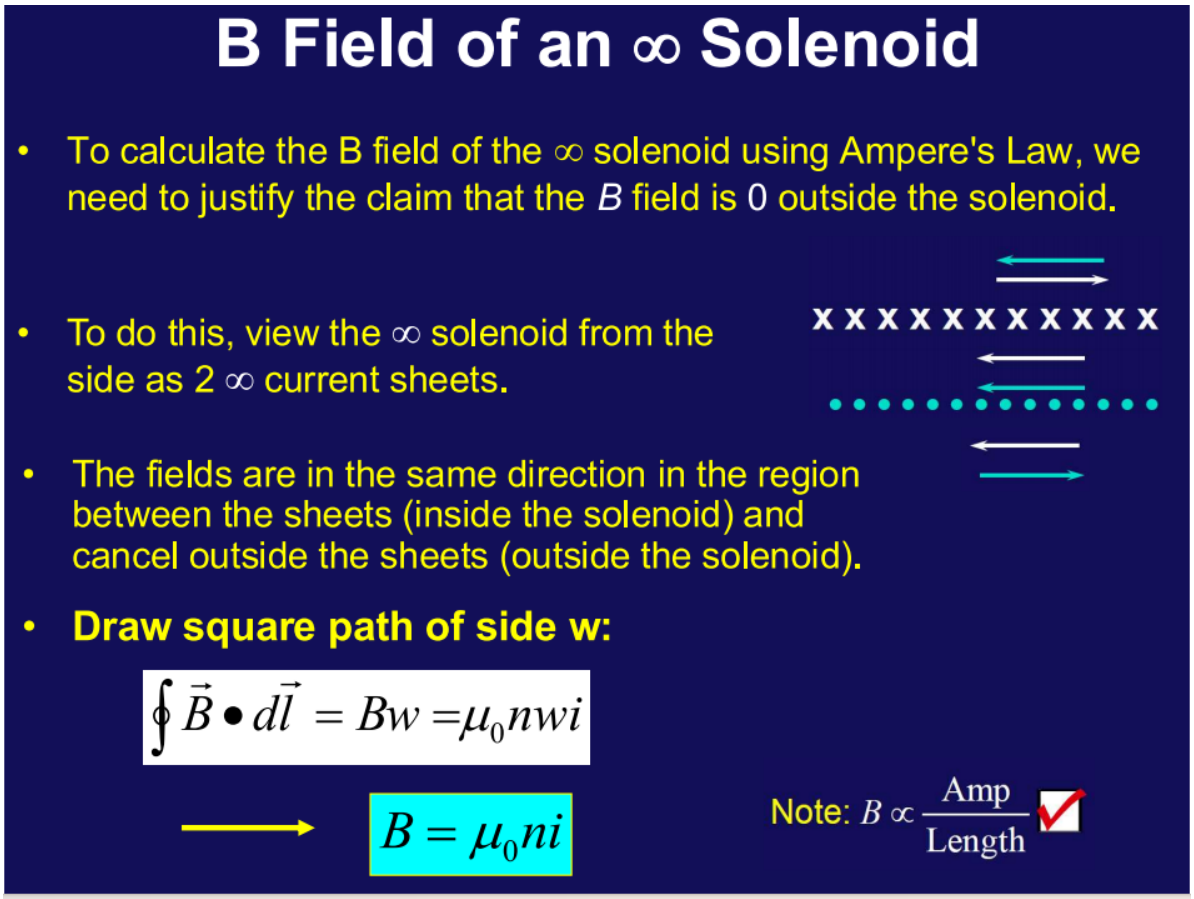


$$B = \frac{1}{2}\mu_0 ni(\cos\beta_1 - \cos\beta_2)$$

如果螺线管无限长

$$B = \mu_0 ni$$

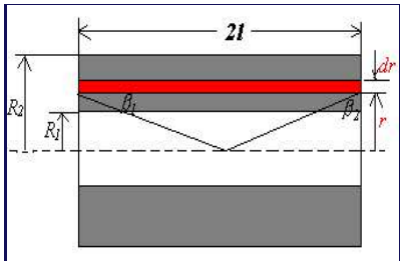
这个也可以用安培环路计算



如果在螺线管一端

$$B = \frac{1}{2}\mu_0 ni$$

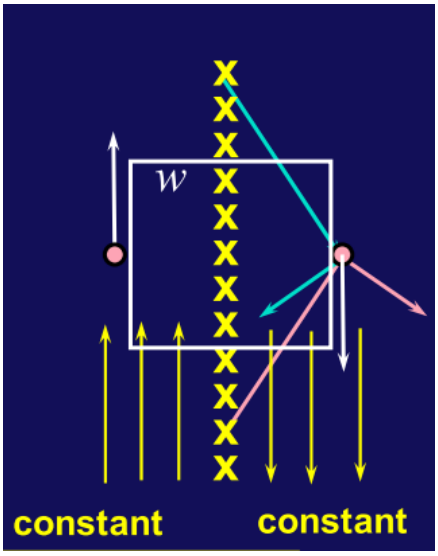
6. 多层螺线管



- 各层加起来的总匝数：N
- $ni = \frac{Ni}{L} \Rightarrow jdr = \frac{Ni}{2l(R_2 - R_1)}dr$

$$B = \mu_0 jl \ln \frac{R_2 + \sqrt{R_2^2 + l^2}}{R_2 + \sqrt{R_2^2 + l^2}}$$

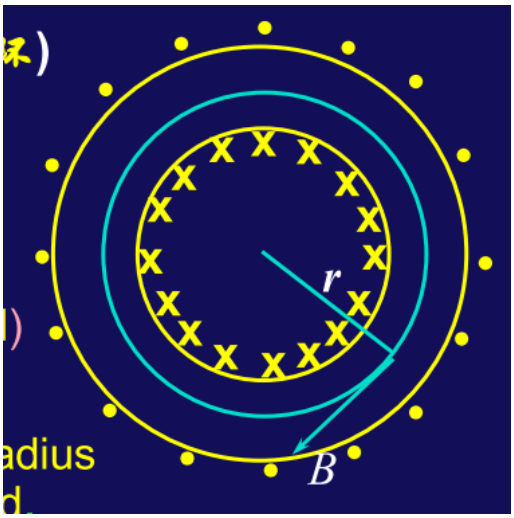
7. 无限大电流板



$$\oint \vec{V} \cdot \vec{dl} = 2Bw = \mu_0 nwi$$

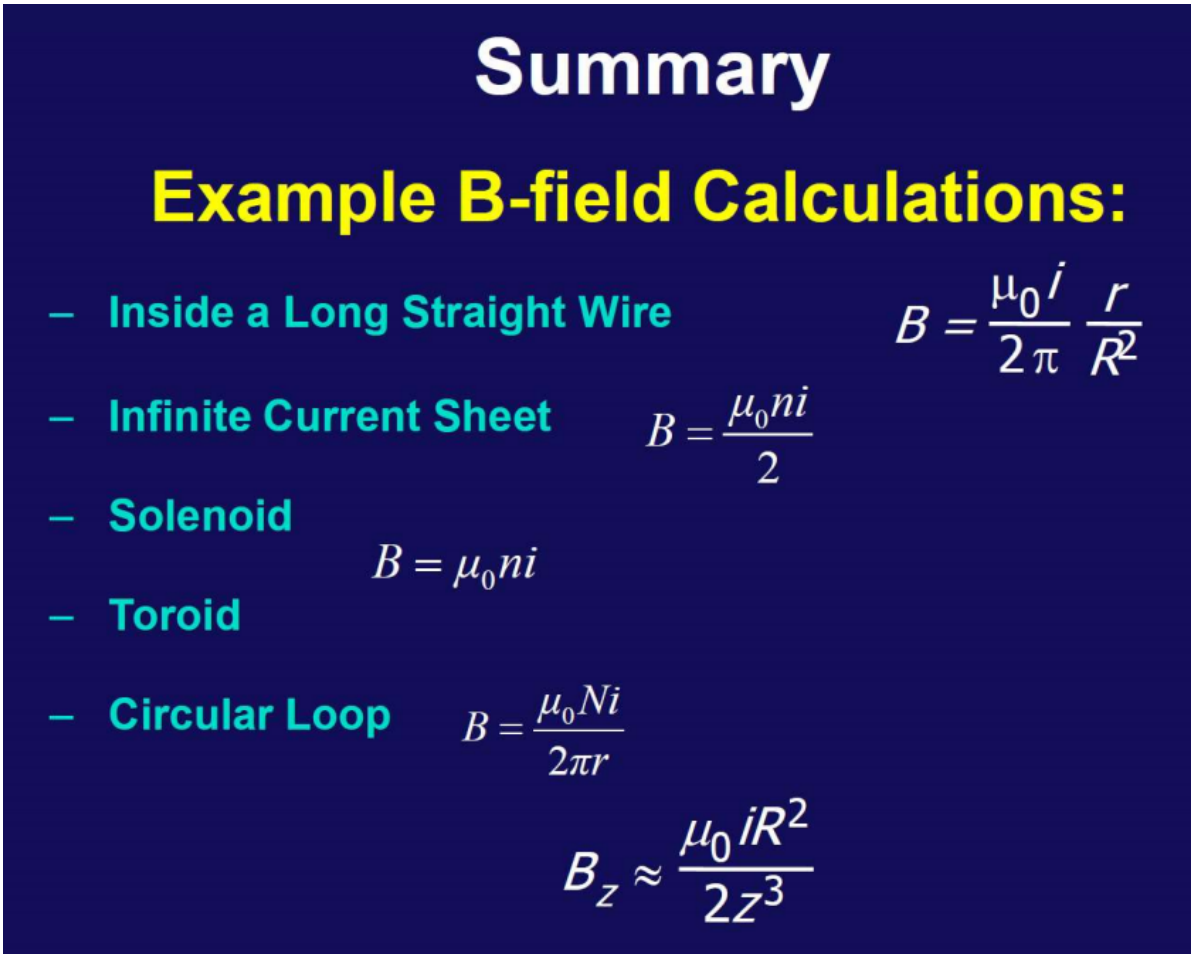
$$B = \frac{1}{2}\mu_0 ni$$

7. 螺绕环(Toroid)



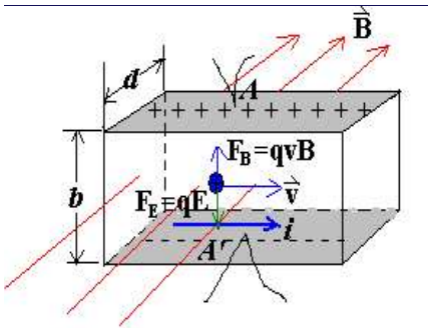
$$\oint \vec{B} \cdot \vec{dl} = \mu_0 Ni$$

$$B = \mu_0 ni$$



8. 质谱仪
9. 瓷瓶
10. 动量仪
11. 回旋加速器

    加速到一定程度后相对论效应明显，需要特别设置磁场
12. 霍尔效应



$$V_{AA'} = \frac{1}{nq}\frac{iB}{d} = \kappa\frac{iB}{d}$$

- Hall Resistance $R_H = \frac{V_{AA'}}{i} = \frac{B}{nqd}$
- 载流子密度 $n = \frac{iB}{qd}\frac{1}{V_{AA'}}$

Chapter 34&35(36) 法拉第电磁感应定律 & 电感(Inductance) 和材料磁性性质

定义

1. 磁通量 $\Phi_B = \iint \vec{B} \cdot d\vec{A}$
2. 磁通匝链数 $\Psi = N\Phi_B$
3. 动生电动势 $\epsilon = \int_{-}^{+} (\vec{v} \times \vec{B}) \cdot \vec{dl}$
    - 绕一端旋转的棒 $\epsilon = -\frac{1}{2} B\omega R^2$
4. 互感系数 $M = \frac{\Psi}{i}$
    - 单位 : 1H=1Wb/1A
    - 互感系数和两线圈自感系数的关系：

$$
\begin{gathered}
No\ flux\ leakage : M = \sqrt{L_1 L_2} \\
Direct\ in\ series : L = L_1 + L_2 + 2M \\
Opposite\ in\ series : L = L_1 + L_2 - 2M
\end{gathered}
$$

5. 自感系数：线圈自己由于电流变化产生感应电动势对自己的影响的衡量 $L = \frac{\Psi}{i}$
    - $\epsilon_L = -\frac{d\Psi}{dt} = -L\frac{di}{dt}$
        - 即使在电阻为0的回路中自感也会限制电流最大值
6. 磁导率permeability constant κ_m
    - 线圈插入磁性材料，$L = \kappa_m L_0$
    - 顺磁逆磁 $\kappa_m \approx 1$ ，铁磁材料 $\kappa_m \approx 10^3 - 10^4$
7. 磁化强度矢量 $\vec{M} = \frac{\sum \vec{\mu_m}}{\Delta V}$
    - μ_m 磁偶极矩
    - 则有 $\oint_l \vec{M} \cdot \vec{dl} = \sum_{inl} i'$
        - i' : induced current 束缚电流
8. 磁场强度 $\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}$
    - 单位：奥斯特 1Os = 1A/m
9. 磁化率susceptibility magnetization coefficient χ_m
    - $\kappa_m = 1 + \chi_m$

定理

1. 法拉第电磁感应定律

$$
\epsilon = -\frac{d\Phi_B}{dt}
$$

$$
\Rightarrow \begin{cases} \oint \vec{E} \cdot \vec{dl} = -\frac{d\Phi_B}{dt} \\ \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \end{cases}
$$

2. 楞次定律

> The induced current will appear in such a direction that it opposes the change in flux that produced it.

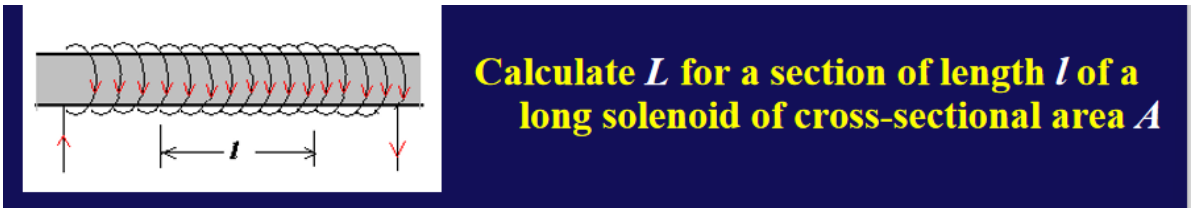
3. **磁性材料加入后的安培环路定律和高斯定律**

$$
\oint_l \vec{H} \cdot \vec{dl} = \sum_{inl} i_0
$$

$$
\oiint \vec{B} \cdot \vec{dA} = 0
$$

模型

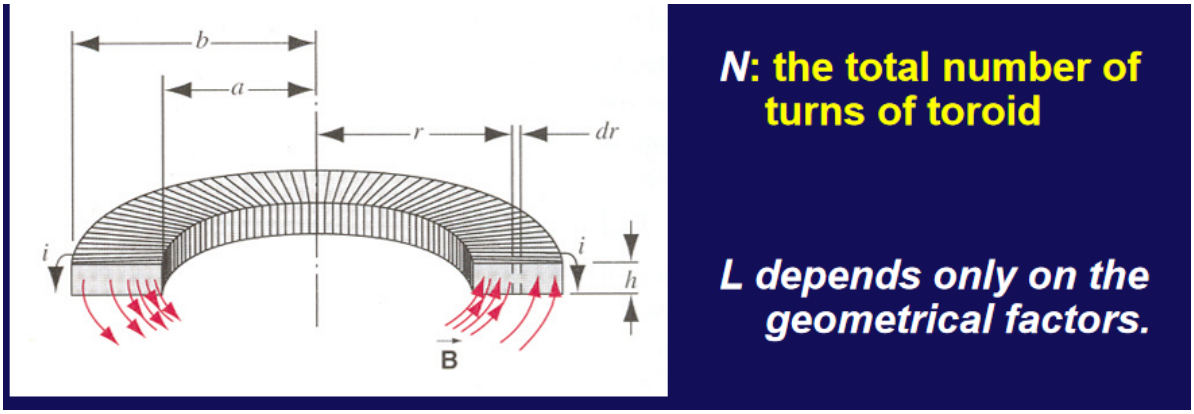
1. 螺线圈的自感系数



$$L = \frac{\Psi}{i} = \frac{nl\mu_0 niA}{i} = \mu_0 n^2 lA$$

$$L_v = \frac{L}{lA} = \mu_0 n^2$$

2. 长方形螺绕环的自感系数

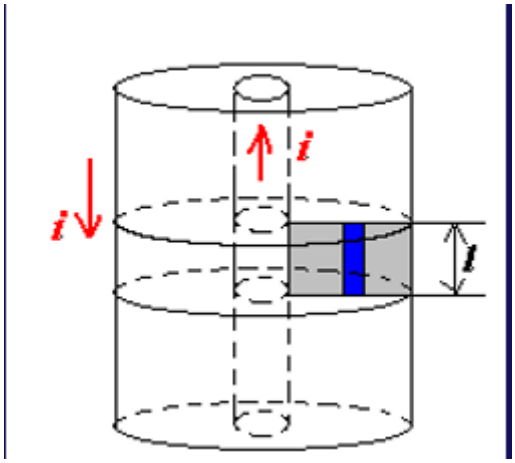


$$\oint B \cdot dl = \mu_0 Ni \Rightarrow B = \frac{\mu_0 iN}{2\pi r}$$

$$\Phi_B = \iint B \cdot dA = \frac{\mu_0 iNh}{2\pi} \ln \frac{b}{a}$$

$$L = \frac{\mu_0 iN^2 h}{2\pi} \ln \frac{b}{a}$$

3. 同轴电缆

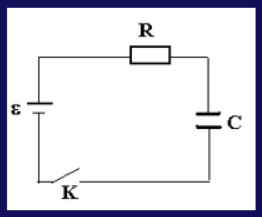


$$B = \frac{\mu_0 i}{2\pi r}$$

$$\Phi_B = \int_{R_1}^{R_2} Bldr = \frac{\mu_0 il}{2\pi r} \ln \frac{R_2}{R_1}$$

35-3 RL Circuits

- **RC circuits:**



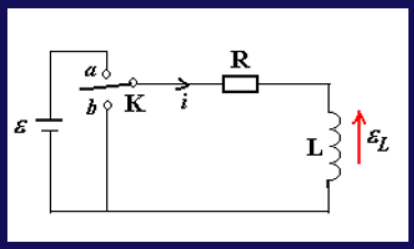
K closed

$$iR+\frac{q}{C}=\varepsilon$$

$$\frac{dq}{dt}+\frac{1}{RC}q=\frac{\varepsilon}{R}$$

$$q=C\varepsilon(1-e^{-t/RC})$$

- **RL circuits:**



K→a

$$iR+L\frac{di}{dt}=\varepsilon$$

$$\frac{di}{dt}=\frac{1}{L}(\varepsilon-iR)=-\frac{R}{L}(i-\frac{\varepsilon}{R})$$

$$i-\frac{\varepsilon}{R}=C'e^{-\frac{R}{L}t}$$

$$t=0,\ i=0,\quad \therefore C'=-\frac{\varepsilon}{R}$$

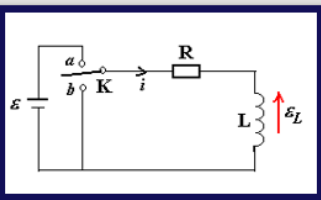
$$i=\frac{\varepsilon}{R}(1-e^{-\frac{R}{L}t})=\frac{\varepsilon}{R}(1-e^{-t/\tau_L})$$

$$\tau_L=\frac{L}{R}$$

$$V_L=-L\frac{di}{dt}=-\varepsilon e^{-t/\tau_L}$$

Induct time constant (时间常数)

RL Circuits (ε on)

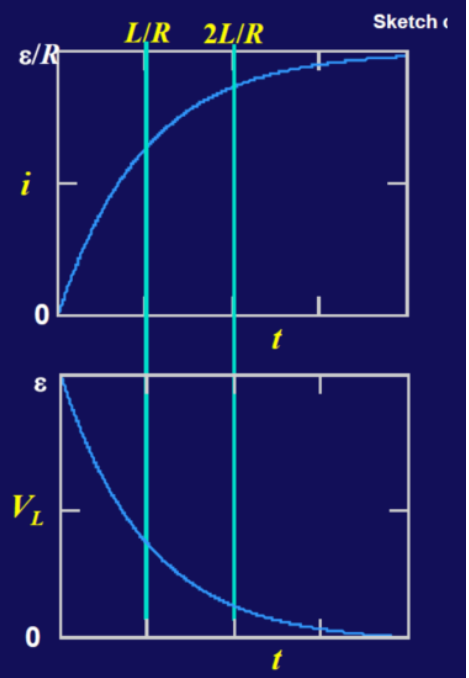


Current:

$$i=\frac{\varepsilon}{R}(1-e^{-Rt/L})$$

Max = ε/R

63% Max at $t = L/R$



Voltage on *L*

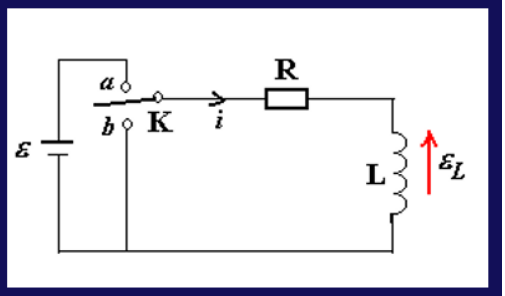
$$V_L=L\frac{di}{dt}=\varepsilon e^{-Rt/L}$$

Max = ε

37% Max at $t = L/R$

RL Circuits

- After the switch has been in position a for a long time, redefined to be t=0, it is moved to position b.



- Loop law:

$$iR + L\frac{di}{dt} = 0; \quad \frac{di}{i} = -\frac{R}{L}dt$$

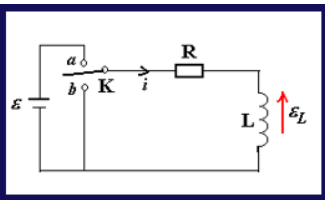
$$i = ce^{-\frac{R}{L}t}$$

- The appropriate initial condition is $i(t=0) = \frac{\varepsilon}{R}$
- The solution then must have the form:

$$i = \frac{\varepsilon}{R}e^{-Rt/L}$$

$$V_L = L\frac{di}{dt} = -\varepsilon e^{-Rt/L}$$

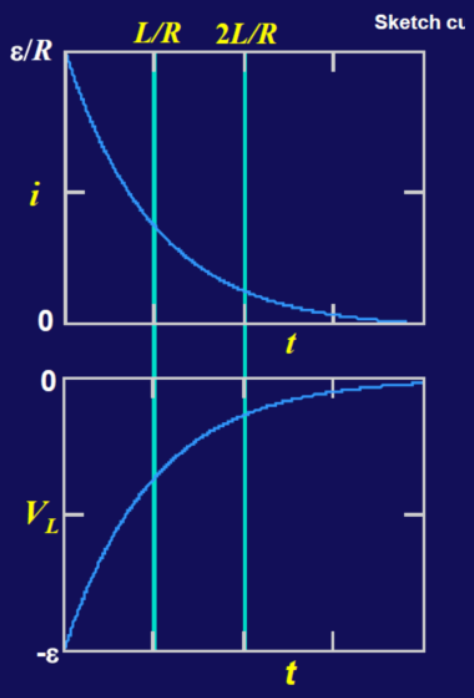
RL Circuits (ε off)



Current: $i = \frac{\varepsilon}{R}e^{-Rt/L}$

Max = ε/R

37% Max at $t = L/R$



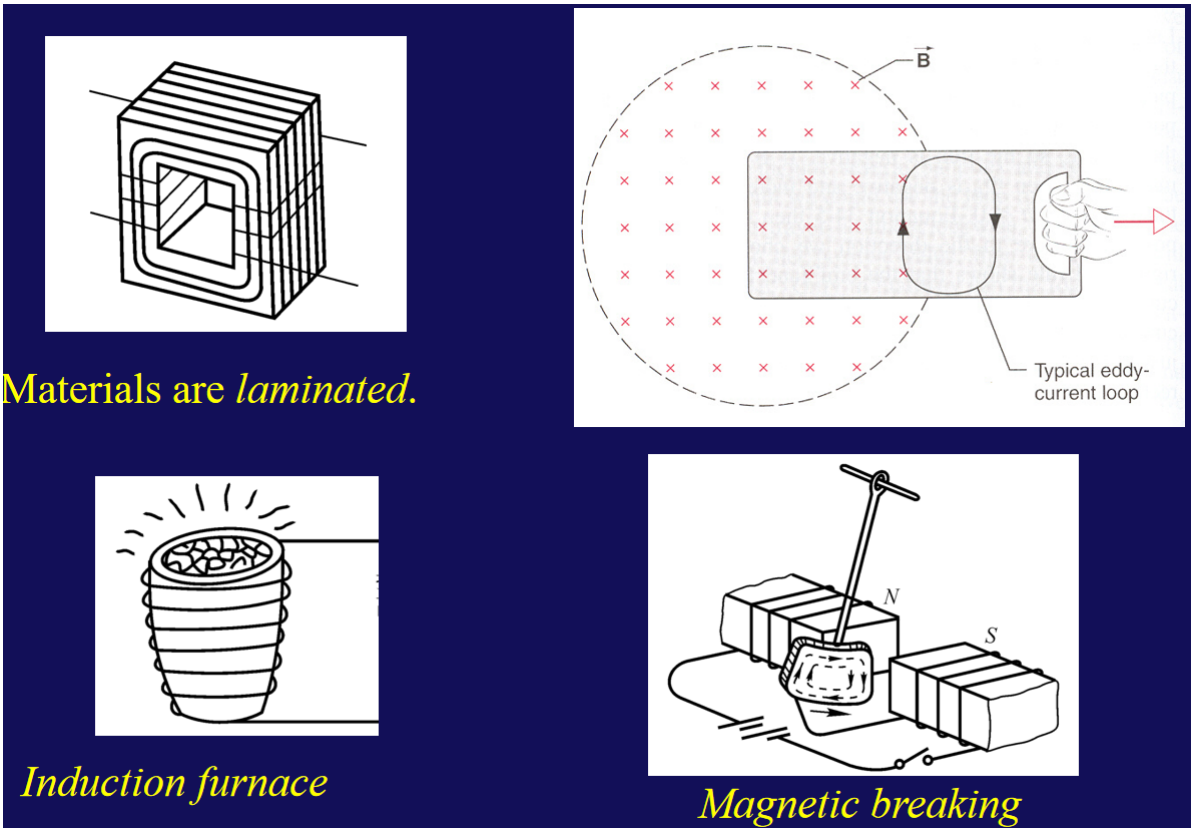
Voltage on *L* $V_L = L\frac{di}{dt} = -\varepsilon e^{-Rt/L}$

Max = -ε

37% Max at $t = L/R$

应用

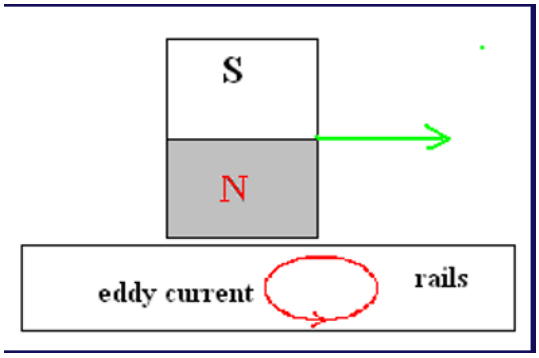
1. 涡流



2. 电磁炮

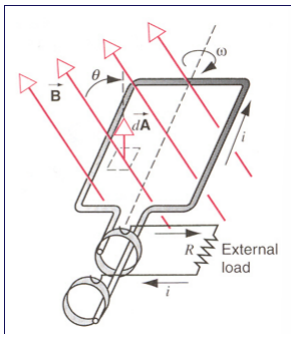
弹丸中感生电动势导致涡流，与线圈的磁场相互作用导致发射

3. 磁悬浮

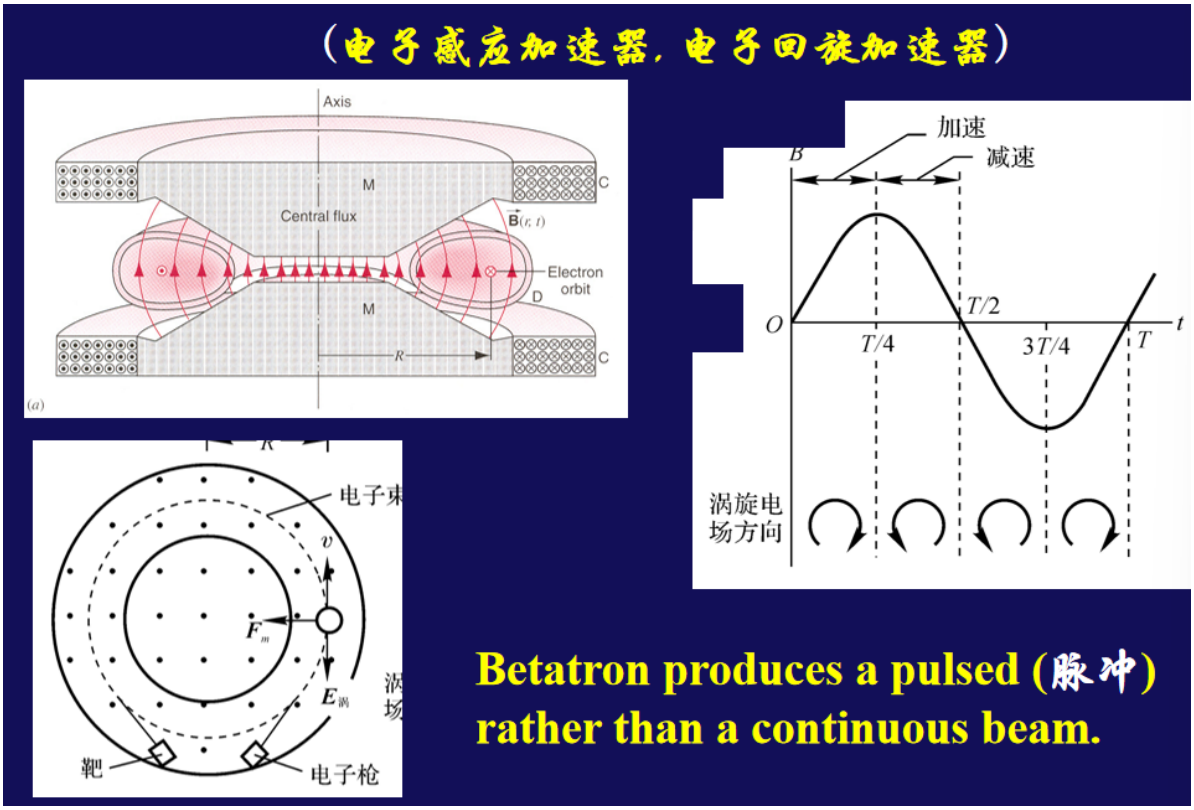


4. 发电机

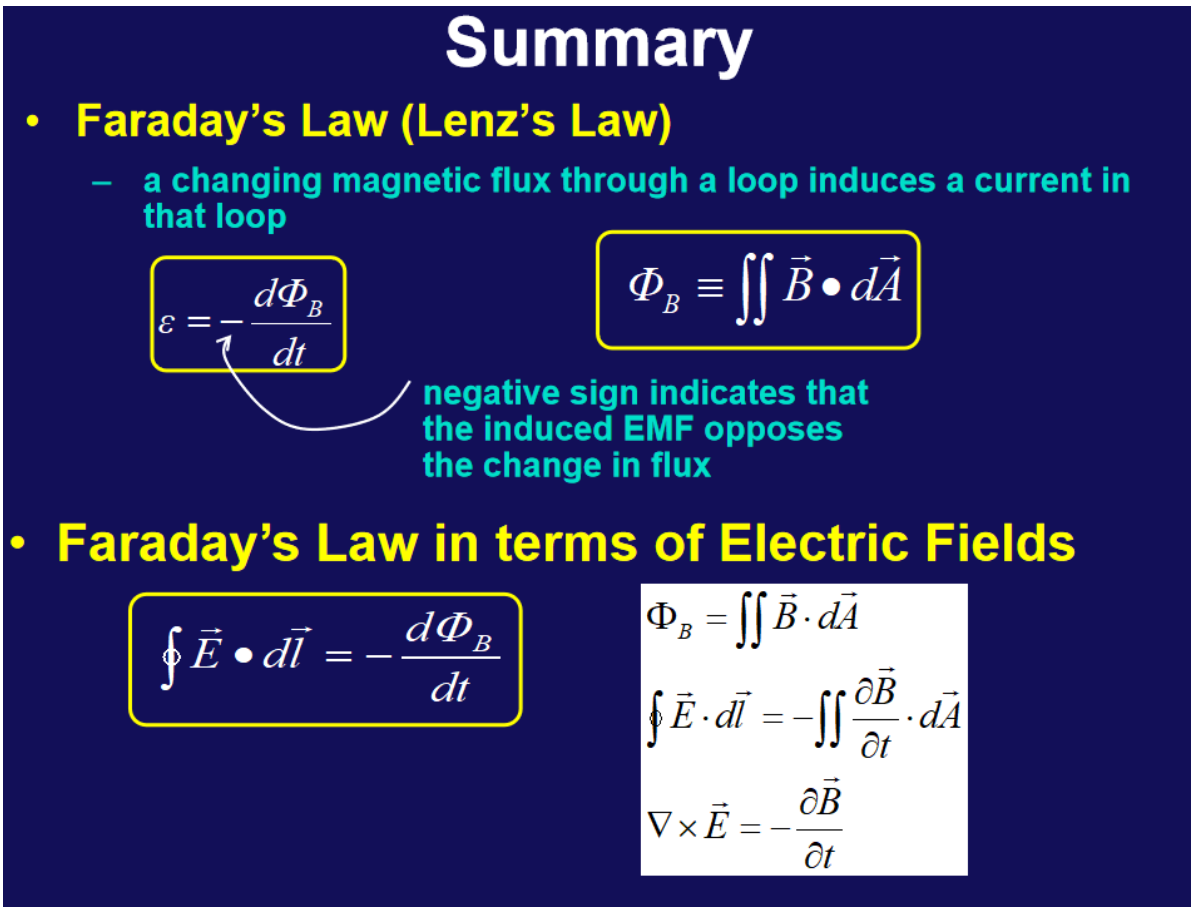
$$\Phi_B = BA\cos\omega t$$
$$\epsilon = BA\omega\sin\omega t$$



5. 电子加速器

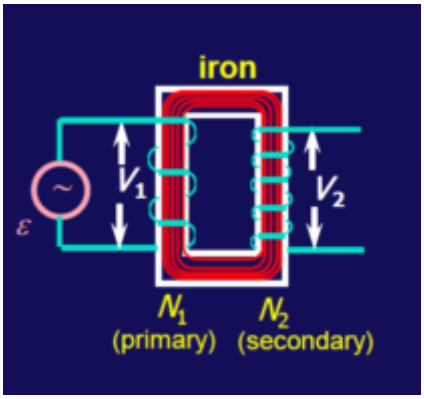


- B 和 v都在变化，最终呈现效果是 R 没变
- 因为会反向增加所以产出不持续



6. 变压器

利用 $M_{12} = M_{21}$ 得到 $V_2 = \frac{N_2}{N_1} V_1$



一些杂七杂八我并看不懂的东西

- 轨道磁矩 $\vec{\mu}_l = -\frac{e}{2m}\vec{L}$
  - 其中 $\vec{L} = \sqrt{L(L+1)}\frac{h}{2\pi}$
  - $\hbar = \frac{h}{2\pi}$
- 自旋磁矩
  - 费米子 $s = \frac{2k+1}{2}\hbar$
  - 玻色子 $s = k\hbar$
- 铁磁材料的磁滞回线
  - 铁磁材料内存在若干分块（磁畴），分块内部磁偶极矩方向相同
  - 施加磁场能让磁偶极矩方向相同，即使退磁后也不能完全恢复到随机化，因而 $B \neq 0$ （称为剩磁）
    - 剩磁小则为软铁磁体，反之为硬铁磁体

总结对比：电与磁

	电	磁
非常重要的常数	$\epsilon_0 \approx 8.86 \times 10^{-12} F/m$	$\mu_0 = 4\pi \times 10^{-7} H/m$
初始安培环路定理	$\oiint \vec{E} \cdot \vec{dA} = \sum q_{in}$	$\oint_l \vec{B} \cdot \vec{dl} = \sum_{inl} i_0$
初始高斯定律	$\oint_l \vec{E} \cdot \vec{dl} = 0$	$\oiint \vec{B} \cdot \vec{dA} = 0$
由于介质引入的新物理量	电感应强度 $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$	磁场强度 $\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}$
	电容 $C = \frac{Q}{V}$	电感 $L = \frac{\Psi}{i}$
引入介质的安培环路定律	\	$\oint_l \vec{H} \cdot \vec{dl} = \sum_{inl} i_0$
引入介质的高斯定律	$\oiint \vec{D} \cdot \vec{dA} = \sum q_{in}$	\
χ	极化率 χ_e	磁化率 χ_m
χ 的引入	对于一般的具有各向同性的材料，有 $P = \chi_e \epsilon_0 E$	对于磁性材料 $\vec{M} = \chi_m \vec{H}$
κ 的引入	插入介质后 $C = \kappa_e C_0$	插入磁性材料后 $L = \kappa_m L_0$ $\vec{B} = \chi_m \mu_0 \vec{H}$
κ	节点常数 $\kappa_e = 1 + \chi_e$	磁导率 $\kappa_m = 1 + \chi_m$
X化强度矢量	极化强度矢量 $\vec{P} = \frac{\sum \vec{p_i}}{\Delta V}$	磁化强度矢量$\vec{M} = \frac{\sum \vec{\mu_m}}{\Delta V}$
X场能量密度	电场能量密度 $u_E = \frac{1}{2}\epsilon_0 E^2 = \frac{1}{2}\vec{D} \cdot \vec{E}$	磁场能量密度 $u_B = \frac{1}{2}\frac{B^2}{\mu_0} = \frac{1}{2}\vec{B} \cdot \vec{H}$